

ZQUAS F1 ENGINE

One Engine. Everything AML. Nothing Shared.

A single GPU-native platform that replaces the entire AML technology stack, from sanctions screening to cross-institutional federation, running locally at each bank with zero data leaving the institution.

What It Replaces

CURRENT STACK	VENDOR(S)	ZQUAS REPLACEMENT
Sanctions screening	Fircosoft, Dow Jones, Refinitiv	GPU policy evaluation (56-policy catalogue, 500,000 entities under 2 seconds)
PEP / adverse media	World-Check, LexisNexis	Integrated in policy engine (CPL rules, 0.5% PEP threshold)
Transaction monitoring	Actimize, Norkom, SAS, Sentinels	Real-time GPU evaluation (not batch, under 10ms alert lifecycle)
Name / entity matching	Fircosoft, Quantexa	GPU identity resolution (canonical IDs, diacritic-aware)
Case management	Actimize, custom	Operator Console (6 roles, investigation workspace, SAR filing)
Suspicious Activity Reporting	Manual / custom	Constitutional governance (agents recommend, MLRO approves)
Network analytics	Quantexa, SynapseGrid	GPU graph propagation (cross-bank contagion detection)
Cross-bank detection	TMNL (winding down)	Privacy-preserving federation (ECDH-PSI + Garbled Circuits)
Regulatory reporting	Custom / Excel	Cryptographic proof per decision (Ed25519-signed, regulator verifies offline)

All of the above runs on a single GPU at each bank. One deployment. One codebase. One team to maintain. Total VRAM: under 1 GB.

The Pipeline

Every entity at the bank passes through six stages. The full pipeline executes in under 2 seconds for 500,000 entities.

STAGE	WHAT HAPPENS	OUTPUT
1. Ingestion	CSV, API, or real-time feed. Multilingual column mapping (Dutch, German, French, English).	Entities in semantic knowledge graph
2. Evaluation	Every entity evaluated against the 56-policy catalogue on GPU. 750,000 complex events/second across the full 500,000-entity cycle.	Risk score + verdict per entity per policy
3. Proof	Every evaluation produces an Ed25519-signed cryptographic proof. Independently verifiable.	Proof bundle (Merkle root + signature)
4. Triage	50,000 GPU-resident agents assess every entity in under 10ms. Constitutional gate: no autonomous SARs.	Clear / Monitor / Investigate / Escalate
5. Federation	All entities enter bilateral PSI with partner banks. Under 10 seconds per round. No raw data shared.	Shared entity matches + risk escalations
6. Network Detection	Risk propagates across transaction + identity edges in unified GPU pass. Cross-bank patterns emerge.	Mule networks, layering chains, circular flows

Key Differentiators

- Compliance Policy Language (CPL): Domain-specific language for AML policy definition. Compiled to GPU bytecode with guaranteed termination. A 56-policy logical catalogue across sanctions, PEP screening, transaction monitoring, fraud detection, KYC/KYB, correspondent banking and regulatory reporting. Identical results on CPU and GPU. Hot-reloadable: policy amendments take effect immediately without restart.
- Semantic Knowledge Graph: Multi-domain entity model with per-field provenance tracking. GPU-optimised entity storage. Every fact about every entity has a tracked origin, confidence score, and timestamp.
- Constitutional Agent Governance: 50,000 autonomous AML agents on GPU, each with per-agent resource budgets, Ed25519-signed warrants, and a constitutional gate. Agents recommend actions but cannot file SARs. A human MLRO must approve every SAR. Multiple independent defence layers prevent autonomous SAR filing.

Privacy-Preserving Federation

Real multi-party computation using established cryptographic protocols: Private Set Intersection for entity matching, Garbled Circuits for secure comparison, Oblivious Transfer for key exchange. Semi-honest security model (appropriate for regulated consortiums). AES-256-GCM transport encryption mandatory. Entity count padding hides set sizes. Dual Ed25519 attestation: both parties sign, regulator verifies. GPU-accelerated. Scales from 5 banks to 500+ without code changes.

Cross-Institutional Graph Propagation

When federation identifies shared entities, the engine creates cross-bank identity edges. Risk propagation traverses both transaction and identity edges in a single GPU pass. Risk flows from one bank's transaction network, across the identity bridge, into the other bank's network. Detects layering chains, mule networks, and circular flows that are invisible to any individual bank.

Measured Performance

Conservative thresholds. All figures measured on NVIDIA RTX 5090. Actual measurements consistently exceed published figures.

METRIC	PERFORMANCE
500,000 entities, 56-policy catalogue	667 ms, median of three
Sustained throughput	750,000 complex events/second across the 500K-entity cycle
Alert lifecycle (ingestion to triage)	Under 10 milliseconds
Federation per bilateral round	Under 10 seconds
Agent triage (50,000 entities)	Under 10 milliseconds
Total GPU memory	Under 1 GB
Maturity	TRL 6, synthetic data, no production deployments
Scaling	Linear (2x entities = 2x time)

Verification & Auditability

CAPABILITY	DETAIL
Cryptographic proof per decision	Ed25519-signed proof bundle with Merkle root. Regulator verifies offline with public key only.
Every primitive RFC/NIST verified	Ed25519: RFC 8032. SHA-256: FIPS 180-4. Blake3: official reference vectors.
Continuous safety verification	Static analysis across entire codebase. Pre-commit gate enforces invariants on every change.
Semi-honest MPC security	Based on ECDLP. Transport: AES-256-GCM. Mutual authentication on every federation round.
Regulatory submissions	DNB InnovationHub: under review. FCA Digital Sandbox: submitted.

Operator Console

Web-based investigation workspace. Six roles: Analyst, Investigator, MLRO, Compliance Officer, Administrator, Regulator. Each role sees only what CPL access policies permit. Bloomberg/Linear dark-theme aesthetic.

FEATURE	WHAT IT DOES
Intelligence Surface	Real-time KPIs, risk distribution, alert feed
Alert Queue	Keyboard navigable (J/K/Enter), ranked by composite risk
Investigation Workspace	Entity profile, counterparty graph, risk timeline, AI-generated narrative, proof panel, provenance badges
SAR Filing	Viewport attestation, typology selection, proof hash. Constitutional gate: MLRO-only.
Federation Demo	Configure peers, run MPC round, view escalations
Regulator Portal	Aggregate view only. No individual entity data. Proof verification with governance public key.
Policy Management	View/branch/amend CPL policies. Dual-approve for constitutional amendments.
Internationalisation	en-GB + nl-NL (188 translation keys, 1:1 parity)

At a Glance
750K

complex events/sec

667 ms

500K entities, 56-policy catalogue

<10ms

alert lifecycle

<10s

federation per round

<1 GB

total VRAM

TRL 6

synthetic data only

All numbers measured on NVIDIA RTX 5090. Medians of 3 runs, 10 warmup iterations. Every result signed.

Danny de Gier **Founder & Engineer**

18 years in financial crime compliance at Tier-1 banks and fintechs. Senior roles at Deutsche Bank, HSBC, RBS, ABN AMRO, ClearBank, Vivid Money. Board Member, Statutory Director, Group CRCO, Group DPO. Professional Postgraduate Diploma in Financial Crime Compliance (International Compliance Association / University of Manchester). Direct regulatory engagement: FCA, FED, DNB, independent monitors.

Self-taught GPU systems programmer. C++23, CUDA (sm_86/89/100/120), Vulkan. Built the entire ZQUAS engine as a solo developer: policy language compiler, GPU adjudication kernel, MPC protocol implementation, cryptographic attestation pipeline, adversarial testing framework, entity resolution graph, rendering engine.

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The architecture is ready. The regulatory window is open. Let's talk.